

Skills Summary

Multiplication: Rows, Arrays, and the Total

Use this reference while completing your worksheet or when studying.

Skill 1 — Finding the total of an array

Rule: rows $\times$ number in each row = total.

The total counts *everything* in the array, so it is usually bigger than either number you multiply.

Example: An array has 4 rows with 6 dots in each row. How many dots in all?

Step 1. The question asks for the total, so multiply the rows by the number in each row.

Step 2. $4 \times 6 = 24$. Skip counting by 6 checks it: 6, 12, 18, 24.

$\Rightarrow$ **24** dots

Try it: An array has 3 rows with 7 dots in each row. How many dots in all?

check: 21

Skill 2 — Equal groups in a story

Rule: number of groups $\times$ number in each group = total.

Bags, boxes, and shelves in a story work just like rows in an array.

Example: There are 5 bags with 8 marbles in each bag. How many marbles in all?

Step 1. The story means 5 equal groups of 8, so the total needs multiplication.

Step 2. $5 \times 8 = 40$.

$\Rightarrow$ **40** marbles

Try it: There are 6 boxes with 4 pencils in each box. How many pencils in all?

check: 24

Skill 3 — Fixing a rows-versus-total mix-up

Check yourself: Does my answer count *everything*?

Adding the rows to the number in each row mixes two different counts — it never gives the total. If your answer is one of the numbers from the problem (or their sum), look again.

Example: A student wrote $5 + 6 = 11$ for an array of 5 rows with 6 dots in each row. Fix the mistake.

Step 1. The sum 11 mixes the row count with the dots in one row — it does not count all five rows of dots, so multiply instead.

Step 2. $5 \times 6 = 30$.

$\Rightarrow$ **30** dots, not 11

Try it: A student wrote $2 + 9 = 11$ for an array of 2 rows with 9 dots in each row. What is the real total?

check: 18

(!) Watch out: The two numbers in an array problem name the *rows* and the *number in each row*. Neither one is the total by itself — the total comes from multiplying them.